

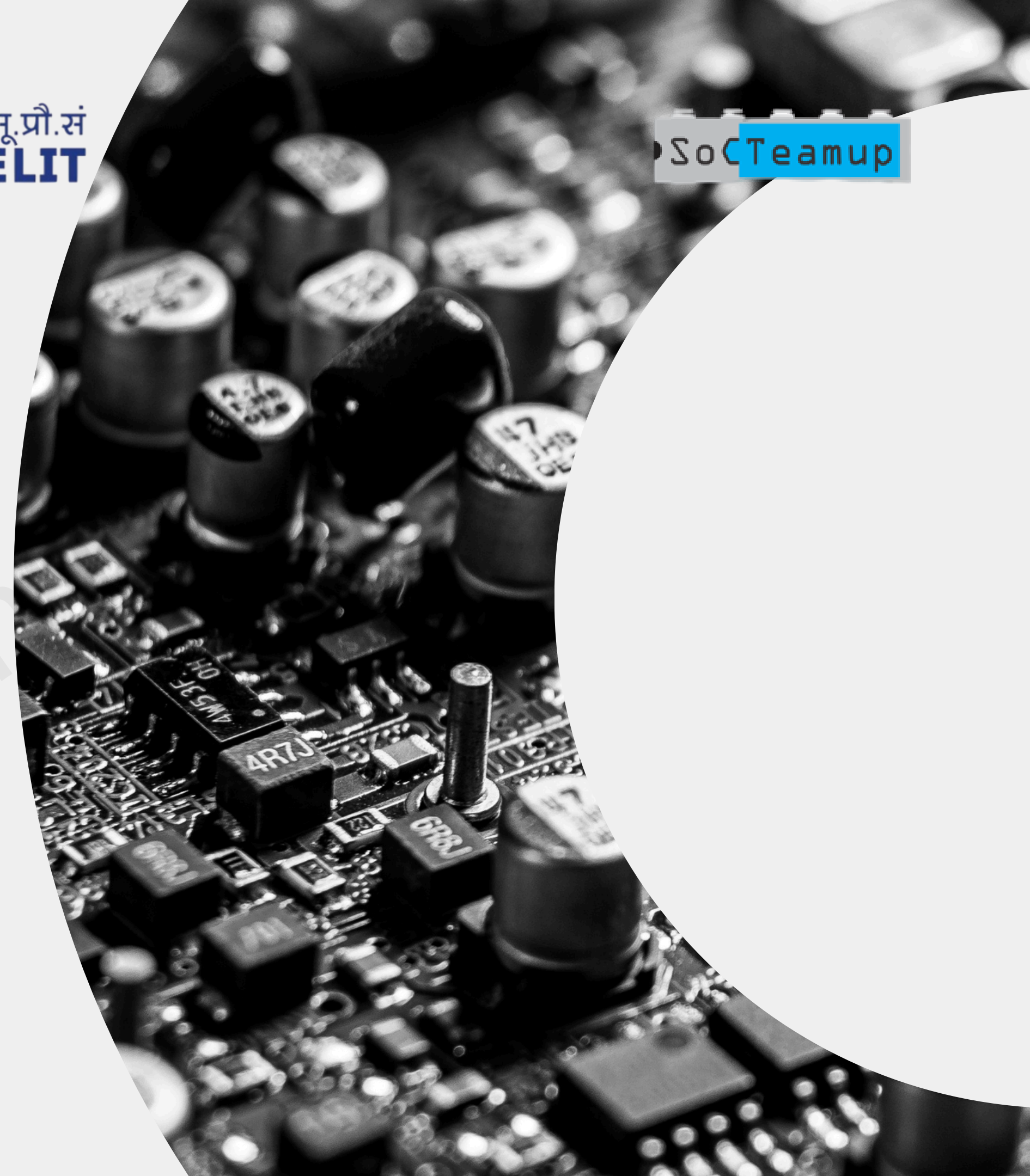


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SoC Teamup

Introduction to parasitics RC delay and wire capacitance





Outline

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➔ What Are Parasitics?

Parasitics :

- Parasitics are unintended electrical effects introduced by layout
- They appear due to wires, vias, and spacing between nets
- Not present in RTL or synthesis — occur after placement and routing

➔ What Are Parasitics?

Types of Parasitics

Parasitic Resistance (R)

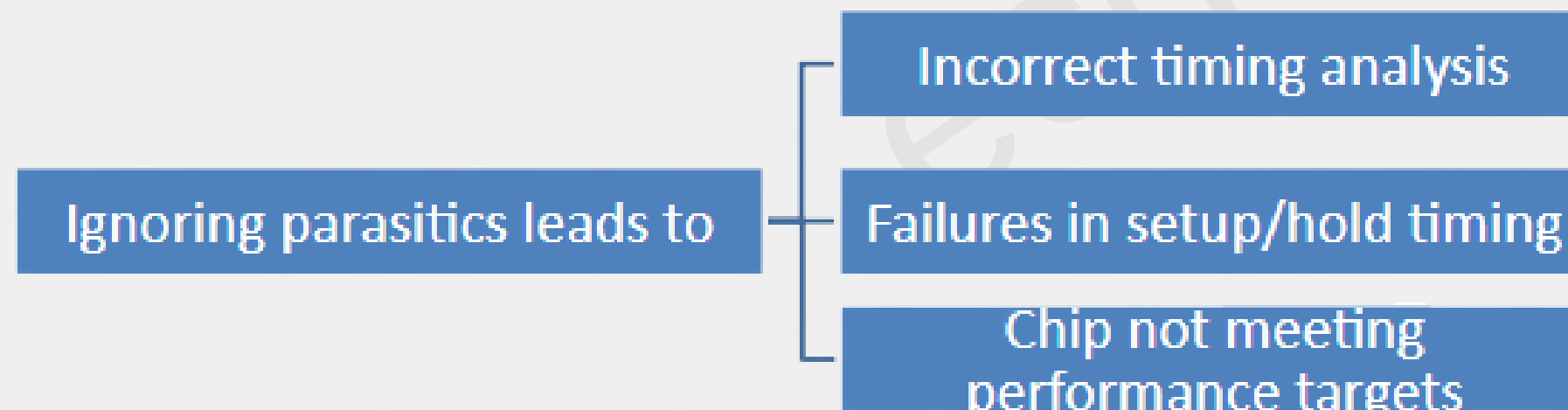
- Caused by long or narrow metal routes
- Via resistance also contributes
- Increases signal delay and IR drop

Parasitic Capacitance (C)

- Wire-to-ground capacitance
- Coupling capacitance between adjacent nets
- Slows transitions and increases power

➔ Why Parasitics Matter in SoC Design

- Parasitics determine whether a routed SoC meets timing, power, and reliability requirements in real silicon.



Why Parasitics Matter in SoC Design

- Increase interconnect delay → setup timing impact
- Cause crosstalk and noise → hold and SI issues
- Raise effective capacitance → higher dynamic power
- Add resistance in power paths → IR drop & EM risk
- Critical for accurate post-layout STA and signoff

RC Delay – The Basics

- Caused by interconnect resistance (R) and capacitance (C)
- Wire delay increases with longer length and smaller width
- Approximated as $\text{Delay} \propto R \times C$
- Dominates over gate delay in deep-submicron SoC designs
- Primary contributor to post-layout timing

RC Delay – The Basics

RC delay = Resistance $\times$ Capacitance

For a wire:

- Resistance depends on length, width, material
- Capacitance depends on wire area and spacing

Time delay (τ) = $R \times C$ affects signal propagation delay.

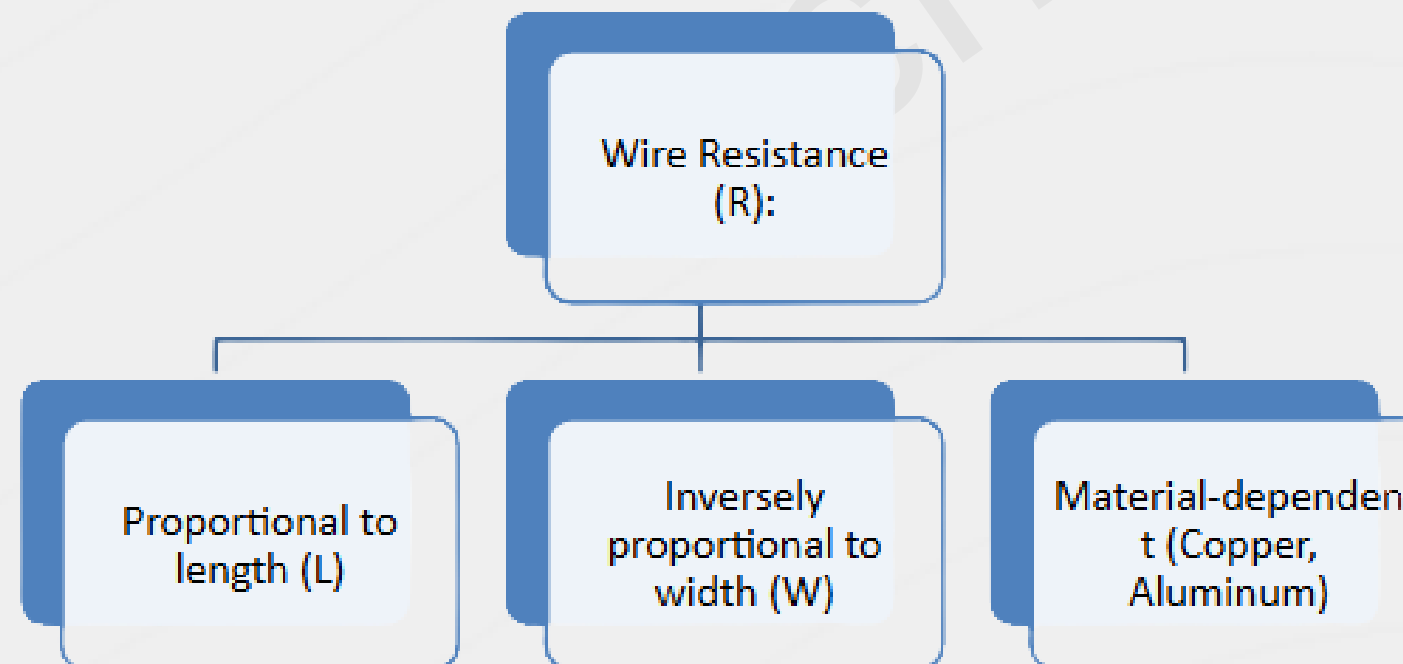
Physical Sources of Resistance (R)

- Metal interconnect resistance due to finite conductivity
- Wire length: longer routes means higher resistance
- Metal width & thickness: narrower/thinner wires means higher resistance
- Temperature effects: higher temperature → higher resistance

➔ Where Does Resistance Come From?

$$R = \rho \times (L / A)$$

Where, ρ = resistivity
 A = cross-sectional area



➔ Physical Sources of Capacitance (C)

- Wire-to-substrate capacitance between metal and silicon
- Coupling capacitance between adjacent signal wires
- Load capacitance from input pins of connected cells
- Via and metal overlap capacitance across layers

➔ Physical Sources of Capacitance (C)

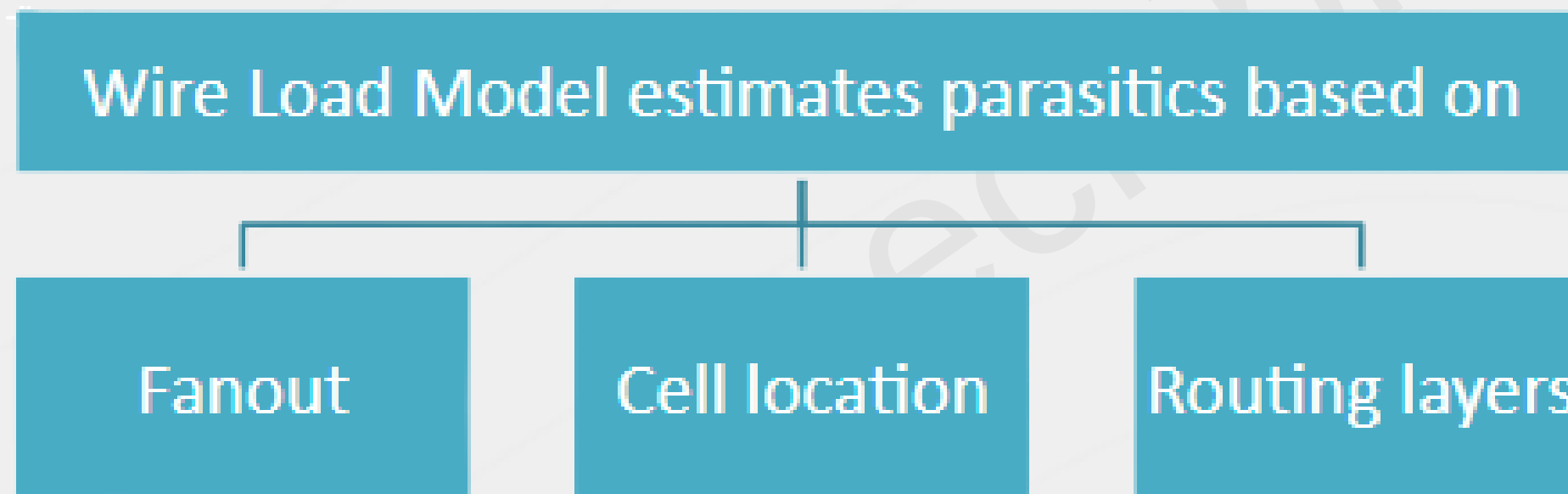
Capacitance arises due to:

- Metal-to-substrate distance (ground capacitance)
- Adjacent metal wires (coupling capacitance)
- Gate input loading

More area = more capacitance
Closer spacing = higher coupling

Wire Load Model

- A wire load model is a pre-layout estimation method used during synthesis to approximate interconnect delay and capacitance based on fanout before actual routing.



- Used before detailed routing to estimate delays.
- Modern flows replace WLM with extracted parasitics post-route.

➔ Lumped vs. Distributed RC Models

Lumped RC	Distributed RC
All R and C placed at a single point	R and C split across segments
Simplified, faster calculation and Less accurate for long wires	More realistic and accurate

Delay Estimation Using RC

- Signal delay is primarily due to interconnect resistance (R) and capacitance (C).
- Delay increases with wire length and capacitance loading.
- Approximated using RC-based delay models (e.g., Elmore delay).
- Post-layout delay = gate delay + RC wire delay.

Delay Estimation Using RC

RC delay approximated as:

$$\tau = 0.69 \times R \times C \text{ (Elmore delay)}$$

More accurate methods use

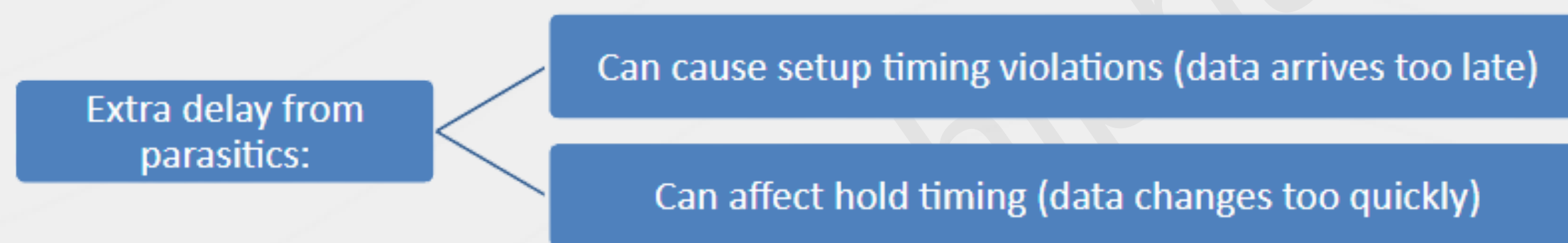
π or T models

SPICE simulations

➔ Parasitic Impact on Setup and Hold Timing

- Parasitics increase data path delay, impacting setup timing
- Excessive RC delay can cause late data arrival → setup violation
- Coupling capacitance can speed up or slow down data, affecting hold timing

➔ Parasitic Impact on Setup and Hold Timing



Hence, accurate parasitic extraction is critical for STA.

Parasitics After Routing

Parasitics are introduced:

- After floorplanning and cell placement
 - Increase significantly after routing
 - Fully known after parasitic extraction (PEX)
 - Used during post-layout STA and signoff
- Parasitics are introduced.

Tools like OpenLane extract parasitics from routed layout to update timing analysis.

➔ Crosstalk and Coupling Capacitance

When two wires run parallel:

They share coupling capacitance

Crosstalk and Coupling Capacitance

Crosstalk occurs when switching activity on one wire affects a neighbour, causing:

Signal delay

Glitches

Noise violations

Coupling Capacitance and Crosstalk

- Coupling capacitance exists between adjacent signal wires
- Switching on one net can induce noise on a neighbouring net
- Causes crosstalk delay (slows down or speeds up signals)
- Can lead to setup or hold timing violations
- More severe in dense routing and advanced technology nodes

Common File Formats: **.spef** and **.sdf**

Common parasitic file
formats:

.spef (Standard Parasitic Exchange
Format)

.sdf (Standard Delay Format) —
contains delays with parasitics

- Used by STA tools for signoff analysis.

➔ Gate Delay vs. Interconnect Delay

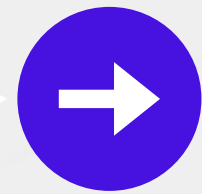
Gate Delay:

- The time taken by a logic cell to switch its output due to transistor behavior.

Interconnect Delay

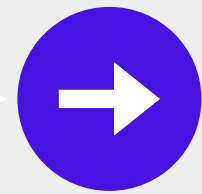
- The time taken by a signal to propagate through wires due to RC parasitic

Hence, RC parasitics are the dominant factor in performance!



Gate Delay vs. Interconnect Delay

Feature	Gate Delay	Interconnect Delay
Source Depends On	Transistor switching Cell size, load, V_t	Wire RC parasitics Wire length, width, spacing
Dominance	Older nodes	Advanced nodes
Modeled By	.lib timing models	SPEF / extracted RC



Summary and Best Practices

Summary

RC parasitics arise due to wire resistance and capacitance

Delay increases with longer and thinner wires

Coupling capacitance leads to crosstalk and glitches

Best Practices

Use parasitic extraction tools post-route

Always include RC parasitics during static timing analysis

Optimize layout to reduce long nets and fanout

Thank You

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